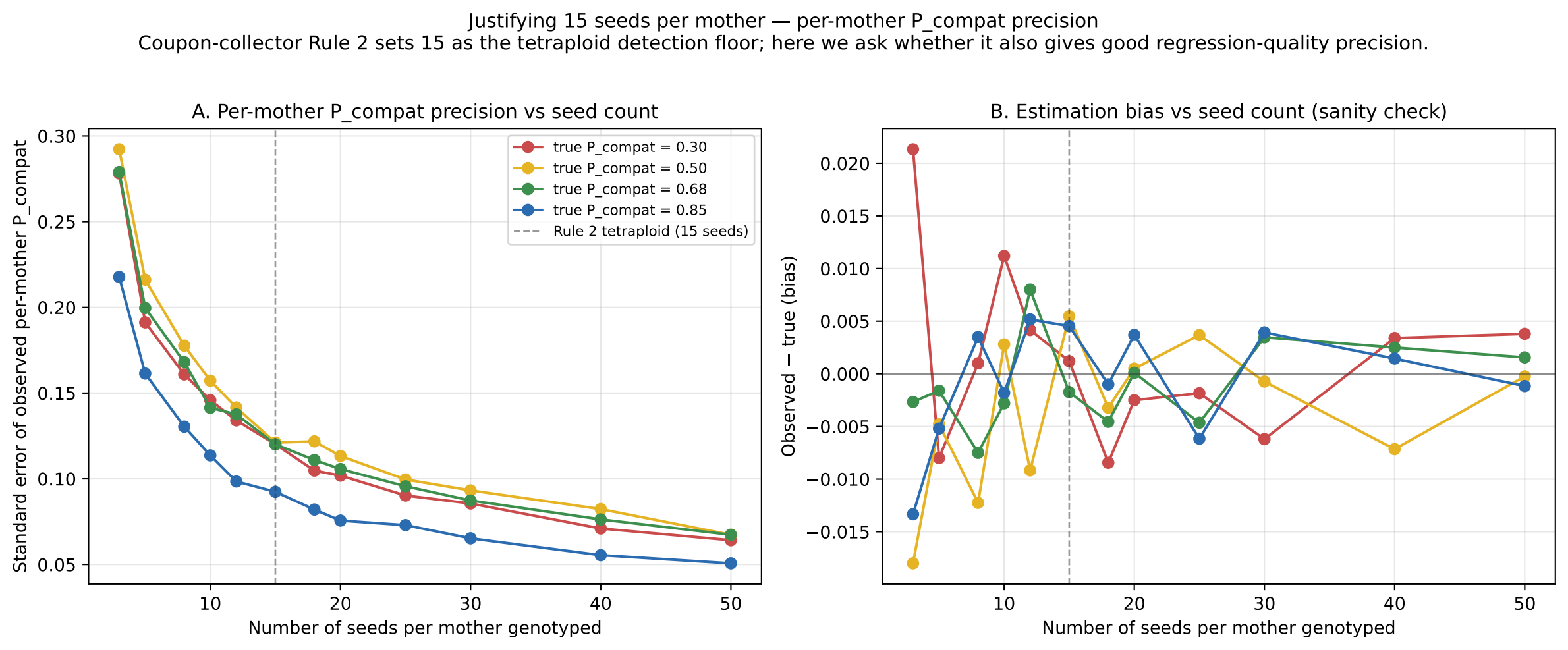
Justifying 15 seeds per mother — per-mother P_compat precision
Coupon-collector Rule 2 sets 15 as the tetraploid detection floor; here we ask whether it also gives good regression-quality precision.
A. Per-mother P_compat precision vs seed count
Standard error of observed per-mother P_compat
Number of seeds per mother genotyped
true P_compat = 0.30
true P_compat = 0.50
true P_compat = 0.68
true P_compat = 0.85
Rule 2 tetraploid (15 seeds)
B. Estimation bias vs seed count (sanity check)
Observed − true (bias)
Number of seeds per mother genotyped